Tayeb Rahman

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Education

Princeton University Princeton, NJ

Computer Science, B.S.E *Expected May 2029*

- *Relevant Coursework*: Programming Systems; Data Structures and Algorithms; Intro CS; Multivariable Calculus; Discrete Math, Probability, and Computer Theory; Linear Algebra; Mechanics; Electricity and Magnetism
- *Honors and Programs*: *Jane Street* Focus Program (**1 of 51** selected nationally), Questbridge Match Scholar

Experience

Hoagie September 2026 - Present

Software Engineer

- Eliminated course conflicts for **5,000+** users by engineering a Django-based scheduling validation layer across 8 semesters and **100+** possible section combinations to ensure zero overlap in 4-year plans
- Contributing to a database in **PostgreSQL** managing **2,000+ courses** and **20+ entity types**, using indexing and optimized queries to ensure fast performance as the course catalog grows
- Decreased manual planning by **20%** by developing a recommendation engine that identifies critical degree gaps by cross-referencing user profiles against **100+** nested requirements, ranking courses by graduation priority

PrincetonBuilds Cohort July 2025 – August 2025

Software Engineering Fellow

- Built and shipped an iOS MVP that generates personalized running routes, onboarded **15+** beta users, and iterated weekly based on retention and engagement metrics
- Increased 7-day retention from **32%** to **61%** across beta cohorts by creating local leaderboards, performance tracking, and Core Data persistence
- Collaborated with *Y Combinator* founders and Princeton Student Ventures in weekly tech reviews to harden architecture, API contracts, and release cadence

Research

Princeton Data and Intelligent Systems September 2026 - Present

Undergraduate AI Researcher

- Developing a resource-aware LLM agent using **Qwen3.8-27B** and *Unsloth* to predict token requirements and dynamically adapt task implementation scope and output fidelity under constrained inference budgets
- Engineered an evaluation pipeline for **5,000** diverse coding, mathematical reasoning, and natural language tasks, integrating tokenizer-level telemetry and structured JSON data processing to quantify token prediction error, budget utilization, and task completion quality
- Designing controlled experiments to benchmark budget-aware planning against fixed-scope generation, investigating tradeoffs between token efficiency, solution completeness, and output quality across varying generation limits

Projects

Cluster Search and Term Mapping - Chrome Extension | *TypeScript, WXT*

- Built a published extension that searches up to **20** input terms and their synonyms, identifying clusters of related concepts across webpages and reducing lookup time
- Implemented an *Aho-Corasick* matcher running in **O(text + matches)**, scanning ∼**2000** characters in ∼**2 ms**
- Combined a **30000**-word offline thesaurus, inflection variants, and cached online lookups with rate limits
- Ranked proximity clusters using distinct-term coverage, match count, and density, enabling one-click navigation to relevant webpage regions

Technical Skills

Languages: Python, Java, C, TypeScript, JavaScript, Swift
Frameworks: Next.js, SwiftUI, Django
Databases & Tools: PostgreSQL, SwiftData, Git, GitHub, OpenCV, MediaPipe, Pandas, MapKit, Unix, Linux
Interests: Chess, Photography, Cinema, One Piece, Minecraft